ALEX ANTONY

Software Engineer | Python Developer | AI/ML Engineer | Full-Stack Developer

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PROFESSIONAL SUMMARY

Final-year MCA student and aspiring Software Engineer with strong Python programming skills and hands-on experience building production-ready applications. Developed an AI Interview Coach (LangChain + FastAPI + React), a YOLOv8-based real-time traffic management system, and a Django-based service management platform. Skilled in software development lifecycle (SDLC), object-oriented programming (OOP), RESTful API design, data structures and algorithms (DSA), and full-stack web development. Strong mathematical foundation (B.Sc. Mathematics) supporting algorithmic thinking and problem solving. Actively solving DSA problems on LeetCode. Seeking a Software Engineer role to contribute to scalable, high-quality software products.

TECHNICAL SKILLS

Programming Languages: Python (primary), JavaScript, TypeScript, Java, C++

Software Development: Object-Oriented Programming (OOP), REST API Design, SDLC, Agile, Unit Testing, Integration Testing, System Testing, Version Control (Git)

Web & Backend Frameworks: FastAPI, Django, Flask, React.js, React Native, HTML, CSS, Bootstrap

AI / ML & GenAI: LangChain, Groq API, Prompt Engineering, YOLOv8 (PyTorch), scikit-learn, NumPy, Pandas, Matplotlib, OpenCV

Databases: MySQL, SQLite, Relational Database Design

Tools & Platforms: Git, GitHub, Linux, Docker (basic), VS Code, Postman

Core Concepts: Data Structures & Algorithms (DSA), Problem Solving, System Design (basics), Statistics, Probability, Linear Algebra

EDUCATION

Master of Computer Applications (MCA) | College of Engineering, Chengannur – APJ Abdul Kalam Technological University | Aug 2024 – Present

Bachelor of Science in Mathematics | SB College, Changanacherry, Kerala | 2020 – 2023 | Score: 70%

PROJECTS

AI Interview Coach | Jun 2026 | GitHub

- Developed a full-stack AI-powered interview preparation web application using Python, LangChain, Groq LLM (LLaMA 3.3 70B), FastAPI, and React.js — demonstrating end-to-end software development and LLM integration skills.
- Designed and implemented RESTful API endpoints using FastAPI for dynamic interview question generation, automated answer evaluation, and session management.
- Built prompt engineering pipeline to generate role-specific technical questions across difficulty levels (Easy/Medium/Hard) covering DSA, System Design, and domain knowledge.
- Implemented automated answer scoring (0–10) with structured feedback — identifying strengths, improvement areas, and providing better answer suggestions.
- Developed session progress tracking with visual score graph across multiple interview attempts, improving user experience and performance monitoring.

Tech Stack: Python, LangChain, Groq API, FastAPI, React.js, REST API

Smart Traffic Management System | Mar 2026 | GitHub

- Designed and implemented a real-time vehicle detection and classification system using YOLOv8 deep learning model, processing live video streams with 90%+ accuracy.
- Built complete data preprocessing pipeline using OpenCV — frame extraction, resizing, normalization, and enhancement for varying lighting and traffic conditions.
- Developed RESTful backend using Flask with model inference and video streaming endpoints; built React.js dashboard for live monitoring and visualization.
- Implemented adaptive traffic signal control algorithm (RED/YELLOW/GREEN phases) driven by real-time vehicle density estimation, reducing manual intervention.
- Conducted comprehensive testing (unit, integration, system) and evaluated model performance using Precision, Recall, F1-score, and mAP metrics.

Tech Stack: Python, YOLOv8 (PyTorch), OpenCV, Flask, React.js, NumPy, Pandas

Autocare – Vehicle Service Management System | 2025 | GitHub

- Developed a multi-role web application (Customer, Mechanic, Manager, Admin) with complete authentication, authorization, and role-based access control (RBAC) using Django.
- Designed normalized relational database schema in MySQL covering complaints, service orders, and real-time status tracking — demonstrating strong database design skills.
- Implemented MVC architecture with modular code structure, following software engineering best practices for maintainability and scalability.

Tech Stack: Django, MySQL, Bootstrap, Python

GameHub – Mobile Game Discovery App | 2026 | GitHub

- Built a cross-platform mobile application using React Native (Expo) with TypeScript targeting both Android and iOS from a single codebase.
- Implemented file-based routing using Expo Router and structured project layout with separate src, assets, and scripts directories following mobile development best practices.
- Demonstrates mobile UI development, TypeScript proficiency, and cross-platform software engineering skills.

Tech Stack: React Native, Expo, TypeScript, JavaScript, Expo Router

LLM & AI Self-Study | Ongoing

- Studying LLM architecture, transformer internals, attention mechanisms, and evolution from GPT to modern instruction-tuned models.
- Exploring Retrieval-Augmented Generation (RAG), LLM agents, tool use, and memory — applied through hands-on projects with LangChain and Groq API.

DSA & Problem Solving | Ongoing

- Actively solving problems on LeetCode covering arrays, strings, recursion, trees, graphs, and dynamic programming.
- Focused on strengthening algorithmic thinking and problem-solving skills for technical interviews at product-based companies.

CERTIFICATIONS

- SWAYAM (NPTEL) – Cloud Computing

AWARDS & ACTIVITIES.

- Actively solving DSA problems on LeetCode — arrays, strings, trees, graphs, dynamic programming — targeting technical interviews at product-based companies.
- Self-studying LLM architecture, RAG systems, LangChain agents, and transformer internals through hands-on projects.